

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

Shanika (192511387)

I Betty A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

20/20
A. J. J.
10/8/20

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1: Video Conferencing

1. Transport Protocol or Application Layer:

- The problem is mainly related to transport-level network conditions and application-level buffering.
- The network has 8% packet loss, which affects audio and video quality.
- 180ms jitter causes irregular packet arrival and communication delay.
- UDP is preferred for real-time communication because it avoids retransmission delays.
- Excessive application buffering can also increase the overall latency.

2. Justification:

- Video conferencing requires low latency for real-time communication.
- It requires low jitter for smooth audio and video.

- Minimum packet loss is needed to avoid distortion and frozen frames.
- UDP provides faster transmission without waiting for acknowledgments.
- The current 8% packet loss and 180 ms jitter significantly affect conferencing quality.

3. Recommended Improvements:

- Use UDP for real-time voice and video transmission
- Implement adaptive jitter buffering
- Use forward error correction (FEC) to reduce the effect of packet loss
- Apply adaptive bitrate according to available bandwidth
- Use QoS to prioritize voice and video traffic.

case study 2 . DNS Infrastructure

1. Reason for High DNS Resolution Time

- The average DNS lookup time is very high at 800ms.
- Users may be accessing DNS servers that are geographically far away.
- DNS server overload can increase response time.
- Lack of effective DNS caching causes repeated time
- Poorly optimized TTL values can result in frequent DNS queries.

2. Proposed DNS Architecture:

- Deploy Anycast DNS servers in multiple geographic regions
- Route users to the nearest DNS server to reduce latency.
- Use DNS caching to store frequently requested records.

- Implement DNS pre-fetching to resolve records before they are needed
- Optimize TTL values to reduce unnecessary DNS lookups and target lookup times below 50ms.

3. Advantages and Limitations:

- Performance: Anycast and caching reduce DNS lookup latency
- Scalability: Multiple DNS servers can handle traffic from users across different regions.
- Availability: Multiple servers provide service even if one server fails
- Limitation: Anycast DNS require more complex configuration and management.
- Incorrect TTL values may cause stale DNS records or frequent lookups.

case study 3 - Distributed DNS solution

1. Impact of centralized DNS Architecture

- All DNS request are sent to a single primary DNS server
- This creates a single point of failure
- Heavy traffic can overload the DNS server and increase lookup time
- The 800ms DNS delay directly affects application performance
- users may experience intermittent delays when accessing cloud services.

2. Distributed DNS solution:

- Deploy multiple DNS servers across different geographic regions.
- use Anycast routing to direct users to the nearest DNS server.

- use DNS caching to reduce repeated queries
- maintain multiple authoritative DNS servers for high availability
- use health monitoring and automatically redirect traffic when a DNS server fails

3. Effect of TTL, Anycast and DNS caching:

- TTL determines how long DNS records remain in a cache.
- Proper TTL values reduce unnecessary DNS queries.
- Anycast routing directs users to a nearby DNS server and reduces latency.
- DNS caching stores previously resolved records and provides faster responses.

case study 4 - TCP Latency

1. TCP - Related Factors

Flow control:

- controls the amount of data sent to the receiver
- small receive window can delay transmission
- increases transaction latency
- Buffering can further increase delay.
- proper buffer settings can improve performance.

Nagle's Algorithm:

- combines small TCP packets
- can delay small trading messages.
- increases acknowledgment latency.
- not ideal for high-frequency trading.

- TCP. NODELAY can reduce the delay

SYN Queue:

- Stores partially established connections
- Heavy traffic can fill the queue
- New connections may be delayed
- Connection establishment time increases.
- Increasing the SYN backlog can help.

2. Effect on Transaction Processing:

- Flow control can delay data transmission
- Nagle's algorithm can delay small messages
- SYN queue limitations delay connections.
- Retransmissions increase latency.
- These factors increase acknowledgment time.

3. Recommended changes

- Disable Nagle's algorithm using

TCP_NODELAY

- Optimize TCP buffer sizes.
- Increase the SYN queue / backlog.
- Use persistent TCP connections
- Reduce TCP retransmissions.

Done ✓